Real Numbers: Mini-Lessons

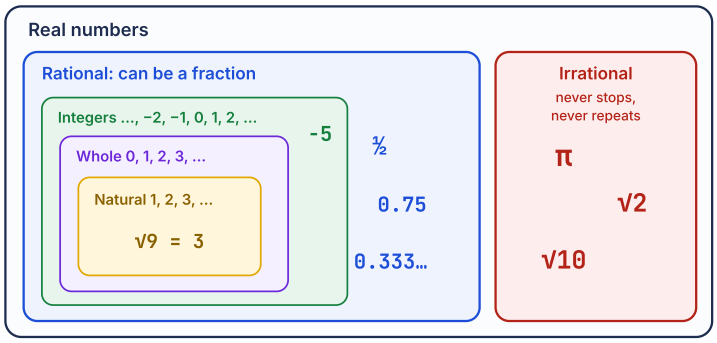
1.1 Rational or irrational? NC.8.NS.1

A number is **rational** if it can be written as a **fraction** of two integers. Its decimal **stops** or **repeats**. A number is **irrational** if its decimal **never stops and never repeats**, like π or √2.

Terminating decimal
a decimal that stops: 0.75

Repeating decimal
a decimal with digits that repeat forever: 0.333...

Perfect square
a whole number times itself: 1, 4, 9, 16, 25, 36, 49, ...



Square roots: √(perfect square) is rational (√49 = 7). √(not a perfect square) is irrational (√10).
Say it this way: "___ is rational because it can be written as a fraction."

Watch out: A long decimal is not always irrational. If it **repeats**, it is rational.

Walk through: `Is √49 rational or irrational?`

1. Is 49 a perfect square?
   Yes: 7 × 7 = 49 → 7 × 7 = 49
2. So what is √49?
   7 → √49 = 7
3. Can 7 be written as a fraction?
   Yes: 7 = 7/1, so √49 is rational → 7 = 7/1 → rational

Check your understanding

1 Score 3 NC.8.NS.1
Is `0.333…` rational or irrational?
(A) **Rational** (B) Irrational

2 Score 3 NC.8.NS.1
Is `√10` rational or irrational?
(A) **Irrational** (B) Rational

3 Score 3 NC.8.NS.1

Which number is **irrational**?

(A) π (B) −5

(C) 0.75 (D) √16

1.2 Repeating decimals to fractions (the 9's rule) NC.8.NS.1

To change a repeating decimal into a fraction: the **repeating digits go on top**. The **bottom is all 9s**, one 9 for each repeating digit. Then **simplify**.

Bar notation

a bar over the digits that repeat: $0.\overline{3}$ = 0.333...

Simplify

divide the top and bottom by the same number

$0.272727\ldots$ $0.\overline{27}$

$$\frac{27}{99} \div 9 = \frac{3}{11}$$

2 repeating digits → two 9s

Decimal	Repeating digits	Fraction	Simplified
$0.\overline{3}$	1 digit	3/9	1/3
$0.\overline{27}$	2 digits	27/99	3/11
$0.\overline{125}$	3 digits	125/999	125/999

Watch out: Count the 9s carefully: one 9 for **each** repeating digit. And always simplify.

Walk through: Write 0.363636… as a fraction

1. Which digits repeat?
   36 → $0.\overline{36}$
2. What is the bottom of the fraction?
   99 (two 9s) → 36/99
3. Simplify 36/99. What number divides both?
   9: 36/99 = 4/11 → 36/99 = 4/11

Check your understanding

1 Score 3 NC.8.NS.1

Write $0.\overline{5}$ as a fraction.

(A) 5/10 (B) **5/9** (C) 1/5 (D) 5/99

2 Score 3 NC.8.NS.1

Write $0.\overline{12}$ as a fraction in simplest form.

(A) 12/9 (B) **4/33** (C) 3/25 (D) 12/99

1.3 Square roots and cube roots NC.8.EE.2

A **square root** undoes squaring: $\sqrt{25} = 5$ because $5 \times 5 = 25$. A **cube root** undoes cubing: $\sqrt[3]{27} = 3$ because $3 \times 3 \times 3 = 27$.

Squared (^{2})

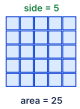
a number times itself: $6^2 = 6 \times 6 = 36$

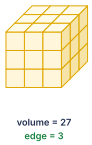
Cubed (^{3})

a number times itself 3 times: $2^3 = 2 \times 2 \times 2 = 8$

$\sqrt{}$ and $\sqrt[3]{}$

square root and cube root: they go backward





n	1	2	3	4	5	6	7	8	9	10
n^2	1	4	9	16	25	36	49	64	81	100
n^3	1	8	27	64	125	216	343	512	729	1000

Watch out: $\sqrt{}$ does **not** mean "divide by 2", and $\sqrt[3]{}$ does **not** mean "divide by 3". $\sqrt{16} = 4$, not 8.

Walk through: Find $\sqrt[3]{125}$

1. What does $\sqrt[3]{125}$ ask?
   What number × itself × itself = 125? → ? × ? × ? = 125
2. Try 5. What is 5 × 5 × 5?
   125 → 5 × 5 × 5 = 125
3. So $\sqrt[3]{125}$ = ?
   5 → $\sqrt[3]{125} = 5$

Check your understanding

1 Score 3 NC.8.EE.2

Find $\sqrt{81}$.

Answer: **9**

2 Score 3 NC.8.EE.2

Find $\sqrt[3]{8}$.

Answer: **2**

3 Score 3 NC.8.NS.1

Which number is a **perfect square**?

(A) 50 (B) 18

(C) 27 (D) **36**

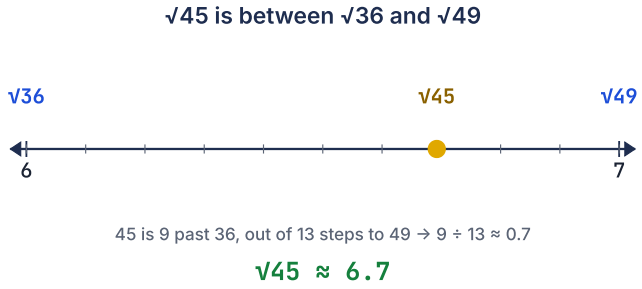
1.4 Estimate roots and order numbers NC.8.NS.2

To estimate a square root that is not a perfect square, find the **perfect squares just below and just above**. The root is **between** their roots. To compare numbers, write them all as **decimals** first.

Estimate
a close guess you can explain

Between
√45 is between 6 and 7 because 36 < 45 < 49

Least to greatest
smallest first, biggest last



4-step estimate of √45: 36 < 45 < 49 → (45 − 36) ÷ (49 − 36) = 9 ÷ 13 ≈ 0.7 → √45 ≈ 6.7

Watch out: Line up the decimals before you compare. 3.5 is bigger than 3.375 even though "375" looks long.

Walk through: `Estimate √20 to the nearest tenth`

1. Which perfect squares are just below and just above 20?
   16 and 25 → 16 < 20 < 25
2. So √20 is between...
   4 and 5 → 4 < √20 < 5
3. Find the tenths: (20 − 16) ÷ (25 − 16) = ?
   4 ÷ 9 ≈ 0.4 → √20 ≈ 4 + 0.4 = 4.4

Check your understanding

1 Score 3 NC.8.NS.2

$\sqrt{30}$ is between which two whole numbers?

- **A** 5 and 6
- B 15 and 16
- C 6 and 7
- D 25 and 36

2 Score 3 NC.8.NS.2

Which is greater: $\sqrt{50}$ or 7.5?

- **A** 7.5
- B √50
- C They are equal

3 Score 3 NC.8.NS.2

Order from least to greatest: $\sqrt{5}, 2.1, 7/3$

- A √5, 2.1, 7/3
- B 2.1, 7/3, √5
- C 7/3, √5, 2.1
- **D** 2.1, √5, 7/3